

01 Introduction to WonderEcho Pro

1. Introduction

WonderEcho Pro, also known as the AI Voice Interaction Box, is equipped with a high-performance noise-canceling microphone and a high-fidelity speaker. Featuring a USB-to-audio module, it offers plug-and-play functionality without the need for drivers and is compatible with multiple systems for both playback and recording.

This device integrates various speech processing modules and uses advanced noise suppression algorithms to effectively filter out background noise, supporting a seamless process from wake-up to speech recognition and interaction. With its modular design, each functional component—such as wake-up, detection, recognition, and synthesis—can be developed and tested independently.

2. Features & Parameters

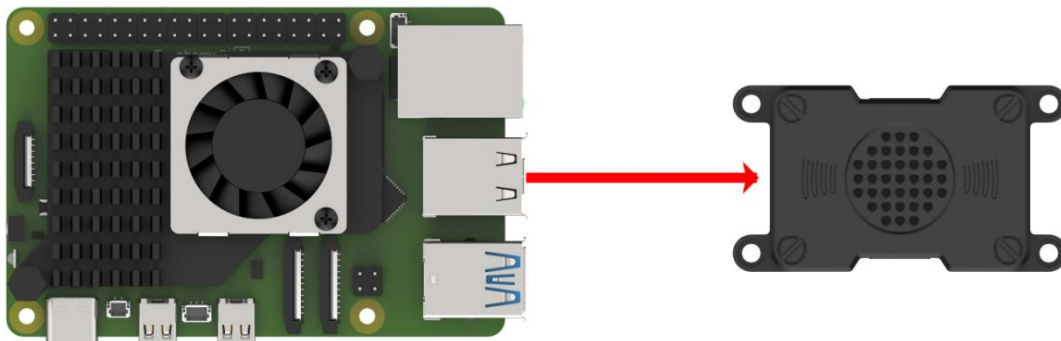
- ① The device is equipped with an onboard microphone and speaker interface, supporting both audio input and output.
- ② It is driver-free, offering seamless plug-and-play functionality with broad system compatibility, including Windows, macOS, Linux, and Android. The "listen/speak" dual function ensures versatility across various platforms.
- ③ It uses a standard USB 2.0 interface.
- ④ Control Interface: USB
- ⑤ Voice Chip Model: CI1302
- ⑥ Speaker Output: 3.0W per channel (4Ω BTL)

- ⑦ Power Supply Voltage: 5V

3. Recording and Playback Test

The following steps demonstrate the process using the Raspberry Pi 5, but the connection and testing procedures are equally applicable to other compatible devices, such as Jetson:

3.1 Connection Diagram and Testing



For Raspberry Pi, you can establish a remote connection to the system using VNC by following the "6 Remote Desktop Tool Installation and Connection / 6.1 Remote Tool Installation and Connection" guide. Once connected, check the top-right corner of the desktop for the microphone and speaker icons, as shown below. If they are present, the connection has been successfully established.



For Jetson-based systems, you can use the Nomachine remote connection tool. Once connected, verify the presence of the speaker icon in the top-right corner of the system.



3.2 Recording Test

- 1) Next, open a new command-line terminal and enter the following command to check for the use of the -l (lowercase L) option, and verify that the card number (card) is 0. This is just an example; please ensure to check your actual query results.

```
arecord -l

> arecord -l
**** List of CAPTURE Hardware Devices ****
card 0: Device [USB PnP Audio Device], device 0: USB Audio [USB Audio]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
```

- 2) Then, use the following command to start recording, replacing the highlighted card number (in red) with the one obtained from the previous query:

```
arecord -D hw:0,0 -f S16_LE -r 16000 -c 2 test.wav
```

- 3) A "test.wav" file will be generated in the current folder.

```
arecord -D hw:0,0 -f S16_LE -r 16000 -c 2 test.wav
```

- 4) You can record for about 5 seconds, then press "Ctrl+C" to stop the recording.

3.3 Playback Test

- 1) After completing the recording, you can check if the file was successfully recorded by running the following command in the current directory:

ls

```
> ls  
test.wav
```

- 2) If "test.wav" appears, the recording was successful. To play the recorded file, use the command below, ensuring you replace the card number with the one obtained from your previous query:

aplay test.wav

```
> aplay test.wav
```